Lecture 4

1. **The hormone that increases the reabsorption of Na^+ in the DCT is:**
   a) Aldosterone
   b) Erythropoietin
   c) Thyroxin
   d) Angiotensin II

Answer: A

2. **The fluid leaving the DCT may be**
   a) Hypotonic or hypertonic but never isotonic
   b) Hypotonic or isotonic but never hypertonic
   c) Hypotonic, isotonic or hypertonic
   d) Isotonic or hypertonic but never hypotonic

Answer: B

3. **Sodium reabsorption occurs in all segments of the nephron except:**
   a) Proximal convoluted tubule.
   b) Distal convoluted tubule.
   c) Descending limb of the loop of Henle.
   d) Ascending limb of the loop of Henle.

Answer: C

4. **The reabsorption of ..... ions and the resulting uptake of water by osmosis is stimulated by the hormone .....**
   a) Sodium, erythropoietin
   b) Sodium, ADH
   c) Sodium, aldosterone
   d) Potassium, aldosterone

Answer: C

5. **The appearance of glucose in the urine**
   a) occurs normally.
   b) indicates the presence of kidney disease.
   c) occurs when the transport carriers for glucose become saturated.
   d) is a result of hypoglycemia.

Answer: C

6. **Which part of the nephron is impermeable to water?**
   a) Proximal convoluted tubule

b) Distal convoluted tubule in the presence of ADH
c) Ascending limb of the loop of Henle
d) Descending limb of the loop of Henle

Answer: C

7. **How does the composition of the filtrate change as it travels through the loop of Henle**?
   a) In the ascending limb, the volume decreases, and in the descending limb, the concentration increases.
   b) In the descending limb, the volume decreases, and in the ascending limb, the concentration decreases.
   c) In the descending limb, the volume decreases, and in the ascending limb, the concentration increases.
   d) In the ascending limb, the volume decreases, and in the descending limb, the concentration decreases.

Answer: C

8. **What is the effect of antidiuretic hormone on the kidney tubules?**
   a) It causes Na^+ to be absorbed from the filtrate into the tubular cells.
   b) It causes the concentration of urine to decrease.
   c) It causes the filtrate volume to increase.
   d) It causes the walls of the collecting duct to become permeable to water.

Answer: D

9. **How does the descending limb of the loop of Henle differ from the ascending limb?**
   a) The descending limb is impermeable to water but permeable to sodium chloride.
   b) The ascending limb is permeable to water but impermeable to sodium chloride.
   c) The descending limb is permeable to water but impermeable to sodium chloride.
   d) The ascending limb is permeable to both water

Answer: C

10. **Under what conditions will the kidney produce concentrated urine? If:**
    a) Glomerular filtration rate is low.
    b) Glomerular filtration rate is high.
    c) Atrial natriuretic peptide is present in blood.

d) Antidiuretic hormone is high in the blood.

Answer: D

11. **In normal kidneys, which of the following is true about the osmolarity of renal tubular fluid that flows through the early distal tubule in the region of the macula densa?**
    a) Usually, isotonic compared with plasma
    b) Usually, hypotonic compared with plasma
    c) Usually, hypertonic compared with plasma
    d) Hypertonic, compared with plasma, in antidiuresis

Answer: B

12. P cells of the collecting tubules are:

a) Less in number than the intercalated cells.
b) Responsible for hydrogen secretion.
c) Responsible for bicarbonate reabsorption.
d) Responsible for sodium and water reabsorption.

Answer: D

13. The U-shaped part of the nephron is:

a) The glomerulus.
b) The Proximal convoluted tubule.
c) The Loop of Henle.
d) The Distal convoluted tubule.

Answer: C

14. Sodium is reabsorbed from all parts of the nephron except:

a) The Distal convoluted tubule.
b) The Proximal convoluted tubule.
c) The descending limb of Loop of Henle.
d) The ascending limb of Loop of Henle.

Answer: C

15. Chloride is reabsorbed actively in which part of the nephron

(a) proximal convoluted tubule
(b) Thick ascending limb of Loop of Henle
(c) descending limb of Loop of Henle
(d) Thin ascending limb of Loop of Henle

Answer: B

16. **Hypertonic urine is excreted due to absorption of water in:**
    a) Collecting ducts
    b) DCT
    c) Ascending part of loop of Henle
    d) Descending part of loop of Henle

Answer: A

17. **A patient is diagnosed with severe "central" diabetes insipidus caused by lack of antidiuretic hormone secretion. Which part of the tubule will have the lowest tubular fluid osmolarity in this patient?**
    a) Proximal tubule
    b) Descending loop of Henle
    c) Distal convoluted tubule
    d) Collecting duct

Answer: C

18. **Which hormone acts on the distal tubule?**
    a) Vasopressin
    b) Thyroid hormone
    c) Corticosterone
    d) Adrenaline

Answer: A

19. **The following hormones acts on DCT of the kidney EXCEPT:**
    a) Angiotensin II.
    b) Parathormone.
    c) ADH.
    d) Aldosterone.

Answer: A

20. **What effect does aldosterone have?**
    a) Increases the reabsorption of Na^+ from the kidney tubules
    b) Makes the kidney tubules more permeable to water
    c) Catalyzes the formation of angiotensin I
    d) Blocks the release of ADH

Answer: A

21. **What is the purpose of ANP in urine production?**
    a) Stimulate the reabsorption of Na^+
    b) Stimulate the reabsorption of Ca^{++}

c) Inhibit the reabsorption of Na^+
d) Stimulate the reabsorption of water

Answer: C

22. **On which of the following does aldosterone exert its greatest effect?**
    a) Glomerulus
    b) Proximal tubule
    c) The loop of Henle
    d) collecting duct

Answer: D

23. **Several hormones regulate the tubular reabsorption of water and electrolytes at different sites in the nephron. Which of the following combination is correct?**
    a) Angiotensin in distal tubule
    b) Aldosterone in collecting ducts
    c) ADH in proximal tubule
    d) ANP in loop of Henle

Answer: B

24. **In the presence of ADH, the filtrate will be isotonic to plasma in the**:
    a) Descending limb of the loop of Henle
    b) Ascending limb of the loop of Henle
    c) Cortical collecting tubule
    d) Medullary collecting tubule
    e) Renal pelvis

Answer: C

25. **Which of the following changes would you expect to find in a dehydrated person deprived of water for 24 hours?**
    a) Decreased plasma renin activity
    b) Decreased plasma antidiuretic hormone concentration
    c) Increased plasma atrial natriuretic peptide concentration
    d) Increased water permeability in the collecting duct

Answer: D

26. Facultative water reabsorption
    a) occurs mostly in the proximal convoluted tubule
    b) represents 87% of renal water reabsorption
    c) depends on ADH

d) requires opening of aquaporin I channels

answer: C

27. **Sodium reabsorbed passively in:**
    a) PCT.
    b) DCT.
    c) descending limb of loop of Henle.
    d) thin part of the ascending limb of loop of Henle.
    e) CD.

answer: D

28. **Which of the following parts of the nephron shows the highest osmolarity:**
    a) PCT.
    b) DCT.
    c) thick part of the ascending limb of loop of Henle.
    d) cortical collecting ducts.
    e) medullary collecting ducts in presence of ADH.

Answer: E

29. **Concerning water reabsorption by the nephrons:**
a) ADH increases water reabsorption from the PCT.
b) the thick part of the ascending limb of loop of Henle is impermeable to water.
c) the collecting ducts are the main sites for water reabsorption.
d) counter current mechanism increases water reabsorption from the DCT.
e) reabsorption of water decreases the osmolarity of urine.

Answer: C

30. **The tubular fluid is iso-osmotic at the end of:**
    a) proximal tubules.
    b) descending limb of loop of Henle.
    c) ascending limb of loop of Henle.
    d) distal tubules.
    e) medullary collecting tubules.

Answer: A

31. **Concerning Henle's loop which of the following is correct:**
    a) the descending limb is impermeable to water.
    b) the cortical type acts as a counter current multiplier.
    c) the juxtamedullary type is shorter than the cortical type.
    d) the descending limb is impermeable to sodium.

Answer: D

32. the renal threshold of glucose is:

a) 100 mg/dl b) 125 mg/dl c) 180 mg/dl d) 375 mg/dl

Answer: C

33. **The plasma concentration at which a substance appears in urine is:**

a) Renal threshold.
b) Load of substance.
c) Transport maximum.
d) Filtration fraction.

Answer: A

34. **The maximum amount of substance that can be transported by renal tubules in 1 minute is:**

a) Renal threshold.
b) Load of substance.
c) Transport maximum.
d) Filtration fraction.

Answer: C

35. **The amount of substance applied to renal tubules in one minute is:**

a) Renal threshold.
b) Load of substance.
c) Transport maximum.
d) Filtration fraction.

Answer: B

36. **The glucose Transport Maximum is:**

a. 80 mg/min
b. 120 mg/min
c. 200 mg/min
d. 375 mg/min

Answer: D

37. **Which of the following is true regarding glucose transport in the renal tubules:**

a. Glucose appears in urine when its plasma concentration is between 120-170 mg/dl.
b. The glucose is not reabsorbed in the renal tubules in diabetic patients.

c. In nephrogenic glucosuria, glucose appears in urine when its plasma concentration reaches 180 mg/dl.
d. Glucose is a normal constituent of urine in healthy subjects.
e. Glucose could appear in the urine before reaching the Transport Maximum.

Answer: E